Team Number: _____

Team Name: _____

Participant names: _____

_____

For Office Use Only:
Part 1: ____ / 37.5
Part 2: ____ / 36
Part 3: ____ / 39
Score: _____ /112.5

Princeton Science Olympiad
Anatomy & Physiology

Please write all your answers here.

Part 1 – Nervous System:

1. __C_____
2. __B_____
3. __D_____
4. __B_____
5. __C_____
6. __C_____
7. __D_____
8. __C_____
9. __A/B/E (1.5 Points for this question, 0.5 point for each)__
10. __A_____
11. __A_____
12. __B_____
13. __E_____
14. __C_____
15. __C_____
16. __E_____
17. __A_____
18. __E_____
19. Total 4 Points

    Any two of the following: epilepsy/seizures, sleep disorders, tumors, encephalopathies, head injuries, stroke, brain death (1 point for correct disorder)

    EEG detects all electrical activity in the brain from the electrodes placed on the scalp of the brain. The frequency pattern and location of electrical activity can be used to diagnose different disorders. (2 Points for description of EEG)

20. Total 5 Points (1 Point each)
    a. _ F; Cerebellum
    b. _ G; Wernicke's Area
    c. _ H; Occipital Lobe
    d. _ A; Frontal lobe
    e. _ D; Medulla oblongata

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21. Total 3 Points (1 Point each)
    a. __ Barbiturates _____
    b. __ Alcohol
    c. __ Caffeine

22. Total 2 Points (1 Point for each of the explanation below)

    binds to and inhibits acetylcholinesterase; prevents breakdown of acetylcholine

23. Total 2 Points
    Arrival of stimulus (needle), Activation of receptor, Action potential in sensory neuron, Information processed in CNS, Activation of a motor neuron, Response by effector (lifts foot)

24. Total 3 points; 1 for name of procedure, 1 for ideal location, 1 for example of disease

    Name of Procedure: Called Lumbar puncture, lumbar tab or spinal tap

    Ideal Location: Fluid removed from subarachnoid space inferior to L3 (inferior to L3 is also acceptable)

    Example of Disease: Most commonly used to confirm suspected meningitis

    Other possible disease answers: subarachnoid hemorrhage, hydrocephalus, benign intracranial hypertension, presence of malignant cells in the CSF, as in carcinomatous meningitis or medulloblastoma

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Part 2 – Sensory System

25. __B_____
26. __C_____
27. __A_____
28. __C_____
29. __ D_____
30. __ B_____
31. __ D_____
32. Total 5 Points (1 Point each)
    a. _ Taste Pore _____
    b. _ Microvilli _____
    c. _ Supporting Cells _____
    d. _ Sensory cells/gustatory receptor cell
    e. _ basal cells _____
33. __C_____
34. __E_____
35. __C_____
36. __C_____
37. 4 Points… 1 point each for sentence…

    Vibration of the basilar membrane causes hair bundles at apex of hair cell to bend back and forth against one another. As the stereocilia bend, the transduction channels are opened, **allowing K+ from the endolymph to enter the hair cell cytosol**. This produces a depolarizing receptor potential, which causes **voltage gated Ca2+** channels to open. **Inflow of Ca2+ triggers** exocytosis of synaptic vesicles containing neurotransmitters.
38. ___C_____
39. __6 Points (1 Point each)_____
    a. __ Semicircular Canals _____
    b. __ Ampulla _____
    c. __ round window _____
    d. __ Utricle _____
    e. __ Oval Window _____
    f. __ Cochlea _____
40. Total 5 Points
    a. (1 point for disease identification, and 1 point for location)
       Otitis Media, an acute infection of the MIDDLE EAR

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b. Bacteria passing into auditory tube from the nasopharynx (1 Point)

c. (1 point for auditory tubes horizontal, and 1 point for why that makes children more susceptible)

   Because their auditory tubes are horizontal, which decreases drainage of fluid from the middle ear.

41. Total 4 points

a. Nyctalopia (1 Point)
b. (1 point for malnutrition/not eat properly, 1 point for Vitamin A mention) Malnutrition, leading to the lack of Vitamin A that is commonly found in fish oils, liver, and dairy products
c. X-linked (1 Point)

Part 3 – Endocrine System

42. __B/D_____(0.5 point for each)_
43. __B_____
44. __C_____
45. __B_____
46. __C_____
47. __B_____
48. __A_____
49. __B_____
50. __C_____
51. __F_____
52. __B_____
53. __A_____
54. __B_____
55. __D_____
56. __G_____
57. __A_____
58. __A_____

Team Number: _____

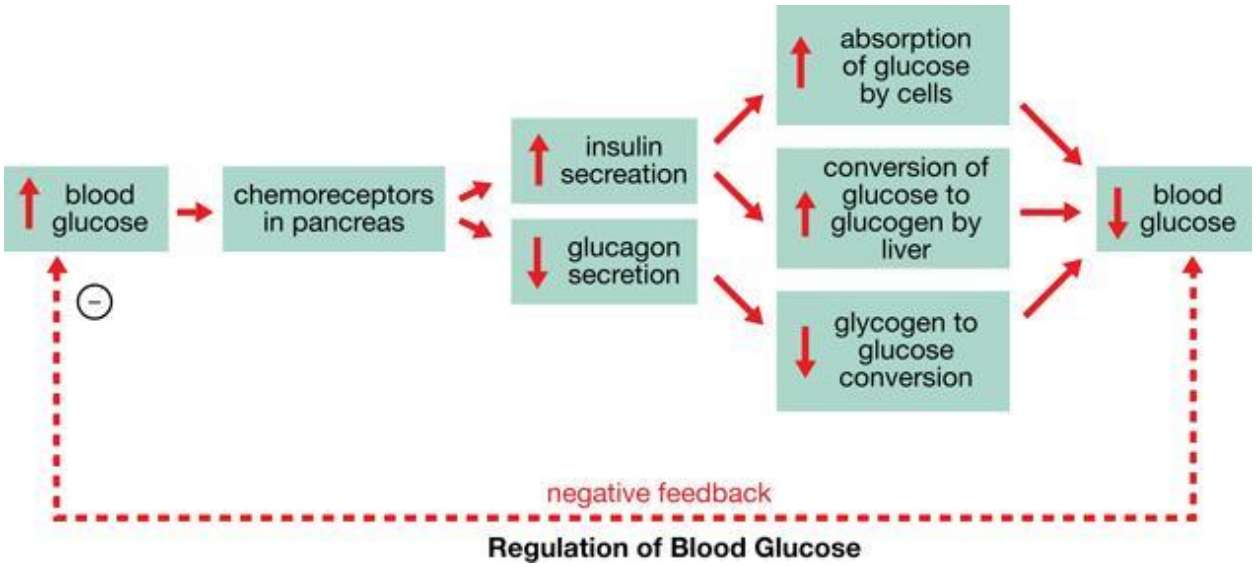
Team Name: _____

Participant names: _____

_____

59. Total 3 Points (1 Point each)
    a. _ Diabetes Mellitus (need to specify diabetes mellitus; 0.5 points if only diabetes)
    b. _ Not enough insulin is being made.
    c. _ Acetone (ketone bodies) have a fruity odor _____
60. Total 5 Points
    a. ____ Arachidonic Acid_ (1 Point)
    b. ____ Red Blood Cells__ (2 Points)
    c. ____ The precursor molecule is derived from phospholipids. (Linoleic acid is also an acceptable answer)__(2 Points)_____

61. Total 6 Points



Regulation of Blood Glucose

- Stimulus: Increase in blood glucose (1 point)
- Increase in insulin secretion from pancreas (1 point)
- Effect:
    - increase in glucose absorption (1 point),
    - increase in conversion of glucose to glucagon in liver (1 point)
- Decrease in blood glucose (1 point)
- Negative Feedback (1 point)

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62. 4 points, 1 point per label
    a. Pars Tuberalis
    b. Pars Distalis
    c. Infundibulum
    d. Posterior Lobe/Neurohypophysis/Posterior Pituitary Gland (Any one of three)

63. Total 4 Points
    a. (1 point for disease identification, 1 point for what it is caused by)

       Grave's Disease, which is caused by antibody production mimicking thyroid-stimulating hormone, resulting in continued stimulation of the thyroid gland and overproduction of the thyroid hormones.

    b. (1 point for overproduction of antibodies, 1 point for similar antibody receptors in extraocular muscle)

       Overproduction of the antibodies may bind to a similar antibody receptors present in the extraocular muscle

Princeton Invitational - Repaired practice edition - Answer key

Division C | Anatomy And Physiology | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

18. (4 points) Authored replacement for source item Question 18: Left foot infection drains via which duct to which vein?

Unit: Central Immune Organs and Lymph Flow.
Worked-case data: A small skin abrasion introduces material into tissue fluid of the left foot. Trace a plausible route toward immune surveillance and return to blood.
Separate investigation observations:
Origin: Antigen enters interstitial fluid around a skin abrasion.
Transport: Afferent lymph carries antigen-bearing cells from the tissue.
Cell encounter: Mature naive lymphocytes meet presented antigen in organized lymphoid tissue.

answer: Thoracic duct to left subclavian vein. (4 points).

33. (4 points) Authored replacement for source item Question 33: Distinguish milk allergy from intolerance in one line each.

Unit: Lactose Intolerance, Obesity, and Exercise Metabolism.
Worked-case data: A teaching model supplies 24 g lactose during transit. In one setting, 18 g can be hydrolyzed; in another, only 6 g can be hydrolyzed.

answer: Allergy: immune hives/wheeze needing avoidance; intolerance: enzymatic bloating/diarrhea managed by dose/lactase. (4 points).